NAME: ______________________ **PERIOD:** ________ **DATE:** ____________

A Percent Needs Its Base

AP Statistics · Unit 1 · Topic 1.3 · B: 2026-09-11 · E: 2026-09-14

[STAR] TODAY'S PROBLEM

Two clubs asked the student council for money to run a Saturday session. Each club surveyed *all* of its members: "Would you attend a Saturday session?" The table shows every response as a count.

At the council meeting, the **Robotics club** presented a **frequency table** (counts) and the **Chess club** presented a **relative-frequency table** (percents). Each club chose the table that made its case look best.

"Would you attend a Saturday session?" (every member surveyed; counts)

Club	Yes	No	Total
Chess	15	3	18
Robotics	27	18	45

FIRST TAKE — before we discuss (5 min)

Which club has more interest in a Saturday session — Chess or Robotics? One sentence and one number.

[BOOK] TABLES FOR A CATEGORICAL VARIABLE (keep on the page)

A **frequency table** gives the count in each category. A **relative-frequency table** gives its proportion or percent: relative frequency $= \dfrac{\text{category count}}{\text{group total}}$. Multiply by 100 for a percent. Counts answer 'how many?'; percents answer 'what share?'. Always identify the group that a percent describes.

Work (a)–(c) from the rules box:

DOK 1 **(a)** How many Robotics members said Yes?

DOK 2 **(b)** Find the percent saying Yes in each club (nearest whole percent). Which is higher?

DOK 3 **(c)** Which club has more student interest? State whether you mean more people or a larger share, and use the matching numbers. Explain why the other measure favors the other club and what a reader needs to know to compare them fairly.

Frame: By more interest, I mean ____________________, so I choose ____________: ________________________.

Frame: The other measure favors ____________ because the club totals are ____________________. Readers also need ____________________.

EXIT — turn this in

One thing the rules box changed about my first take: ____